

Kali Linux Cybersecurity, Firmware & CTF Command Reference

File Reconnaissance

Command:
`file firmware.bin`

Description:
Identifies file type, architecture, and format.

Example:
`file init_v2`

Strings Analysis

Command:
`strings -a firmware.bin`

Description:
Extracts printable strings from binaries.

Example:
`strings firmware.bin | grep -i password`

Hex Inspection

Command:
`xxd firmware.bin | less`

Description:
View raw bytes and offsets.

Example:
`grep -aob "FLAG" firmware.bin`

Firmware Extraction

Command:
`binwalk -Me firmware.bin`

Description:
Detects and extracts embedded filesystems and binaries.

Example:
`binwalk -Me orpheus_fw_v5.bin`

ELF Analysis

Commands:
`readelf -h binary`
`objdump -d binary`
`ldd binary`

Description:
Determine architecture, libraries, and disassembly.

Radare2

Commands:
`r2 binary`
`aaa`
`afl`
`pdf`

Description:
Interactive reverse engineering and disassembly.

Ghidra

Description:
Import binary -> Analyze -> Inspect functions, strings, crypto usage.

Typical searches:
main, strcmp, SHA256, AES, password, flag

SquashFS

Commands:
unsquashfs filesystem.squashfs
mksquashfs squashfs-root new.squashfs

Description:
Extract and rebuild embedded Linux filesystems.

Password Cracking

Commands:
john hash.txt --wordlist=/usr/share/wordlists/rockyou.txt
hashcat -m 1800 hash.txt rockyou.txt

Description:
Recover passwords from shadow hashes.

Packet Analysis

Commands:
tshark -r capture.pcap
tshark -r capture.pcap -T fields -e data

Description:
Inspect packets and extract payloads.

Scapy

```
from scapy.all import *  
pkts = rdpcap("capture.pcap")
```

Description:
Programmatic packet analysis.

AES Decryption Example

```
from Crypto.Cipher import AES  
  
cipher=AES.new(key,AES.MODE_CBC,iv)  
plain=cipher.decrypt(ciphertext)
```

RSA Analysis

```
from Crypto.Util.number import *  
from sympy.ntheory.modular import crt
```

Description:
Used for RSA and CRT attacks.

CTF Workflow

1. file
2. sha256sum
3. strings
4. xxd
5. binwalk
6. extract
7. reverse engineer
8. derive keys
9. decrypt

10. recover flag